UNIVERSITY OF THE
WITWATERSRAND,
JOHANNESBURG

Human Research Ethics Committee: (Medical)
FWA Registered No IRB 00001223

SECRETARIAT: Suite 189, Private Bag x2600, Houghton 2041, South Africa Tel: +27-11-274 9200 Fax: +27-11-274 9281

30 June 2022

EMAILED & COURIERED

Prof MF Chersich,
Wits RHI Head Quarters
Hillbrow Health Precinct
22 Esselen Street
Hillbrow
2001

Email: mchersich@wrhi.ac.za

Dear Prof Chersich,

PROTOCOL: HEAT002 - INNOVATIVE MACHINE LEARNING AND MULTI-SOURCE DATA ANALYSIS TOWARDS DEVELOPMENT OF AN URBAN HEAT-HEALTH EARLY WARNING SYSTEM FOR AFRICAN CITIES

ETHICS REFERENCE NO: 220606

RE : FINAL ETHICS APPROVAL

This is to certify that the above-mentioned trial has been approved by the University of the Witwatersrand, Human Research Ethics Committee (HREC), and the Protocol/Expert Reviewer. Date of Meeting where trial was reviewed: 24 June 2022.

The University of the Witwatersrand, Human Research Ethics Committee Approval Granted for the above mentioned study is valid for five years. Where required by Sponsor to have approval on a more frequent basis it remains the responsibility of the Sponsor and Investigator to apply for continuing review and approval, or for the duration of the Trial.

1. It is the responsibility of the Sponsor and Principal Investigator to ensure, where required, that relevant approvals are in place and compliance with the following is adhered to before a trial may begin:

- If trial is being conducted in Provincial Health facilities: Approval from the Hospital CEO / Clinic Manager / District Research Committee (whichever is applicable) be obtained.
- The study is submitted onto The National Health Research Database (NHRD).
- The relevant approvals are uploaded onto the NHRD system: Ethics Approval, SAHPRA Approval, Hospital CEO / Clinic Manager / District Research Committee Approval.

* A copy of the SAHPRA Approval and/or SAHPRA Notification letter must be submitted to the Ethics Secretariat Office for record purposes (IF SAHPRA APPROVAL / NOTIFICATION IS APPLICABLE).

* The study is conducted according to the protocol submitted to the University of the Witwatersrand, Human Research Ethics Committee. Any amendments to the protocol must first be submitted to the Human Research Ethics Committee for approval.

* During the study, the University of the Witwatersrand, Human Research Ethics Committee is informed immediately of :

- Any Unexpected Serious Adverse Events or Unexpected Adverse Drug Reactions, which, in the Investigator and/or the Sponsor's opinion are suspected to be related to the study drug. (Refer to POL-IEC–001 and SOP-IEC–005, Item 3.4).
- Any data received during the trial which, may cast doubt on the validity of the continuation of the study.

* The University of the Witwatersrand, Human Research Ethics Committee is notified of any decision to discontinue the study and the reason stated.

* The Investigators authorised by this approval participate in this study. Additional Investigators shall be submitted to the University of the Witwatersrand, Human Research Ethics Committee for approval prior to their participation in the study.

* In the event of an authorised Investigator ceasing to participate in the study, the University of the Witwatersrand, Human Research Ethics Committee must be informed and the reason for such cessation given.

2. PRINCIPLES OF INFORMED CONSENT:
* The University of the Witwatersrand, Human Research Ethics Committee requires that in all studies, the Principles of Informed Consent are adhered to. This applies to volunteers as well as patients.

3. PROGRESS REPORTS:
* The University of the Witwatersrand, Human Research Ethics Committee requests that the SAHPRA Progress Reports be submitted twice a year either in March and September or six monthly from start of study to the HREC Secretariat Office - email: EthicsRegulatory@witshealth.co.za and a report of the final results, at the conclusion of the study. (IF APPLICABLE)

4. REIMBURSEMENT TO PATIENTS FOR TRANSPORT:
* The Human Research Ethics Committee: (Medical) is in agreement that reimbursement per visit is according to the South African Health Products Regulatory Authority and that reimbursement should be appropriate according to the situation.

5. TRANSPORT AND STORAGE OF BLOOD AND TISSUE SAMPLES IN SOUTH AFRICA:
* If blood specimens are to be stored for future analysis and is planned that such analysis will be done outside Wits, then the blood must be stored at a facility in South Africa agreed with the relevant IRB, with release of sub-samples only once projects have been approved by the local Research Ethics Committee applicable to where the analysis will be done as well as by the Wits Human Research Ethics Committee: (Medical).

6. GENETIC TESTING:
* The Human Research Ethics Committee: Medical; will not approve open-ended genetic testing as this does not fit the Human Research Ethics Committee criteria.

7. GOOD CLINICAL PRACTICE:
* The South African Department of Health, South African Health Products Regulatory Authority requires Good Clinical Practice (GCP) Training for all Investigators in Clinical Trials, and that GCP training be renewed every three (3) years.

As yet, there are no National Guidelines for the content of GCP courses. Until these are available the Wits Human Research Ethics Committee (Medical) will note courses completed by Investigators without approval of the content of the individual courses.

8. THE SUPPORTING APPROVAL DOCUMENTS ARE ATTACHED:

8.1 Ethics Approval Form signed by the Chairperson of the HREC - Kindly return the copy of the Approval Form signed by the Principal Investigator(s) per email: EthicsRegulatory@witshealth.co.za for our records (this is applicable with the initial Approval).

8.2 List of members present at the HREC meeting held as per INDEPENDENT ETHICS COMMITTEE APPROVAL FORM

9. WE AWAIT YOUR RESPONSES AS REQUESTED: Ensure to have these documents forwarded at the earliest for the HREC records.

* SAHPRA Approval letter and/or letter of Notification before the above study may commence / or where an Amendment may be implemented (IF SAHPRA APPROVAL / NOTIFICATION IS APPLICABLE). It remains the responsibility of the Principal Investigator and/or Sponsor to ensure that the relevant approvals are in place.

* Copy of Independent Ethics Declaration Approval Form signed by the Principal Investigator. (this is applicable with the initial Approval).

* Kindly forward the above to the undersigned to email: EthicsRegulatory@witshealth.co.za at your earliest convenience.

The above has been noted for the Ethics Committee information and records.

KINDLY FORWARD TO THE RELEVANT INVESTIGATORS / CRA / SPONSOR / STUDY CO-ORDINATORS - WHERE APPLICABLE

Regards,

PROF CLEMENT PENNY

For and on behalf of the Human Research Ethics Committee: (Medical)

UNIVERSITY OF THE WITWATERSRAND JOHANNESBURG

INDEPENDENT ETHICS COMMITTEE APPROVAL FORM

Ethics Reference No.	220606	Date of Meeting	24-Jun-2022
		Recertification Due	26 May 2023 (If applicable)

Principal Investigators:	Prof MF Chersich	Investigators:
Co-Investigator	Dr G Maimela (nee Kgosana)	
Co-Principal Investigator	Mr CJ Parker	

Protocol Title:	Innovative machine learning and multi-source data analysis towards development of an urban heat-health Early Warning System for African cities

DOCUMENTS REVIEWED			Tick As Appropriate	Yes	No
Protocol Number	HEAT002	Date:	03-Jun-2022		
Protocol	HEAT002 Protocol, Version 1.0	Date:	03-Jun-2022	☑	☐
Investigator's Brochure	N/A - Version: N/A - Dated:			☑	☐
Subject Information/Consent Form					
Advertisements					
Questionnaires					
Insurance/Compensation	N/A	Valid From:	To:	☐	☑
Synopsis of Study/Trial Summary	Multi-source data analysis of urban heat-health			☑	☐
Other Documentation	Protocol Synopsis - Dated: 03 Jun 2022			☑	☐
	Wits Research Office - HREC Application Form - Dated: 07 Jun 2022			☑	☐
	Notice of Award - Number: 1U54TW012083-01 - Dated: 23 Sep 2021			☑	☐
Relevant Trial Hospital/(s)	Wits RHI - Esselen Street			☑	☐
Syndicate and/or Research Unit	Wits RHI - Research Centre Clinical Research Site			☑	☐

DETAILS OF COMMITTEE

Name	University of the Witwatersrand Human Research Ethics Committee: (Medical)
Address	Research Office, Senate House University of the Witwatersrand, 1 Jan Smuts Avenue, BRAAMFONTEIN, Johannesburg, 2000

DETAILS OF MEETING	Yes	No
Is the Investigator a member of the committee ?	☐	☑
If "Yes" did he/she vote ?	☐	☑
Is the Committee organised and operated according to applicable laws and regulations together with ?	☑	☐
Local GCP requirements ?	☑	☐
ICH GCP requirements ?	☑	☐
FDA GCP requirements ? FWA Registered No. IRB00001223	☑	☐
Progress reports required either in March and September or six monthly from start of study ?	☑	☐

DECISION ON APPROVAL : is approval given to conduct the trial ?	Tick As Appropriate
Yes - with no conditions	☑
Remarks : Our expectation is that our requirements are fulfulled once a company has submitted and uploaded relevant approvals onto NHRD.	
Yes - with conditions	☐
Specify conditions :	
No	☐
Specify reasons	

SIGNATURES

I confirm that the details on this form are correct:		Date
Name: **Prof C Penny** **Chair / Deputy Chair of Committee**	**Signature:** L B Penny	**30 June 2022**

DECLARATION OF INVESTIGATOR/(S)

To be completed and ONE COPY returned to the Secretariat for the HREC at Wits Health Consortium, 31 Princess of Wales Terrace, Parktown, 2193 or Email: EthicsRegulatory@witshealth.co.za

I/We fully understand the conditions under which I am/we are authorised to carry out and complete the above-mentioned research and I/we agree to ensure full compliance with these conditions. Should any amendment, alteration or departure be contemplated from the research procedure methodology or manner of execution, I/we will communicate with the Chairman of the Human Research Ethics Committee: (Medical) for approval prior to acting on any of the above mentioned proposed amendments, alterations or departures. I am/we are fully aware that any unauthosised amendment, alteration or departure as above will amount to misconduct and may lead to the institution of disciplinary procedures.

Any approval given by the HREC is conditional upon consent being obtained by the Investigator/s from the Superintendent (or equivalent official) of the Hospital, Clinic or Institution in which the research is, in part or full, to take place.

The Chairman may of course at his discretion place the matter before the full Committee.

DATE: 08 July 2022 SIGNATURE: _____ NAME: Matthew Francis Chersich

PROTOCOL NUMBER **HEAT002** ETHICS REF.: **220606**

UNIVERSITY OF THE WITWATERSRAND JOHANNESBURG

HUMAN RESEARCH ETHICS COMMITTEE MEMBERS: (MEDICAL) UNIVERSITY OF THE WITWATERSRAND

Attendance Register for the Ethics Meeting held on 24 June 2022 from 12:00 - 15:00

Venue: MEETING TAKING PLACE VIRTUALLY AND FACE-TO-FACE: WHC: 4th Floor Boardroom, 31 Princess of Wales Terrace, Parktown

AFFILIATED TO THE UNIVERSITY OF THE WITWATERSRAND

Surname	Initials	Title	Discipline/s	Academic Qualifications	Gender	Present
Adam	Y	Prof	Obstetrics & Gynaecology	MB BCh; FCOG	F	Present
Badal-Faesen	S	Dr	Infectious Diseases/HIV/TB	MB BCh	F	Absent
Billings	B	Dr	Anatomical Sciences	PhD	M	Absent
Bronstein	VL	Prof	Lawyer	Associate Professor in Law	F	Present
Conradie	FM	Dr	Infectious Diseases/HIV/TB	MB BCh; DTM&H; MSc; Dip HIV Man	F	Absent
Cooper	PA	Prof	Paediatrics	MB BCh, PhD, DCH (SA), FCPaeds (SA)	M	Absent
de Andrade	VM	Dr	Speech Pathology and Audiology	BA; MA; PhD	M	Absent
Demetriou	GS	Prof	Oncology, Haematology, Medicine	MBChB, FCP (SA), Cert Med Oncology	F	Absent
Dhai	A	Prof	Biomedical Ethics	MB ChB; FCOG; LLM; PGDipIntResEthics; PhD	F	Absent
Diana	NE	Dr	Nephrology	MBBCh; Cert Neph (Phys); MMed (Int Med)	F	Present
Etheredge	H	Dr	Biomedical Ethics	MSc Med, BA; PhD	F	Present
Feldman	C	Prof	Pulmonology	MB BCh, PhD, DSc, FCP (SA), FRCP	M	Absent
Gerrand	P	Dr	Social Work	PhD (Social Work)	F	Present
Kerr	R	Dr	Human Genetics	BSc, BScHons, MSc, PhD	F	Present
Lulua	DR	Dr	Health Science Education	MBBCh	F	Absent
Mashegoana	P	Ms	Psychology	BA (Behavioural Science) Psychology, Sociology	F	Present
Mokhachane	M	Dr	Clinical Medicine	MB BCh, FCP (Paeds) SA, MMed, Neonatology (SA)	F	Absent
Naidoo	S	Prof	Public Health	MB BCh, DMTH, DHSM, DOH, MMED	M	Absent
Naran	NH	Dr	Chemical Pathology	PhD	M	Present
Penny	C	Prof	Internal Medicine	BSc Hons, PhD	M	Present
Potterton	J	Prof	Physiotherapy	PhD, MSc Physiotherapy; BSc Physio	F	Absent
Prodehl	LM	Dr	Gastroenterology	MBChB; MMed (Surg); FCS (SA); Cert GIT (Surg)	F	Present
Ross	M	Prof	Public Health	MB ChB, FFCH(SA), FOM(UK)	F	Absent
Sanne	IM	Prof	Infectious Diseases/HIV/TB	MB BCh, FCP (SA), DTM&H; MMed & PhD	M	Absent
Smith	C	Prof	Psychiatry	BA, BA (Hons), M.A (Clin.Psych), PhD	F	Present
Stacey	SL	Dr	Pulmonology	MB BCH, FCP	F	Present
Szabo	CP	Prof	Psychiatry	MB BCh, MMed, MScMed, PhD; FCPsych(SA)	M	Absent
Thom	RGM	Prof	Psychiatry	MB ChB, DCH, FCPsych, PhD	F	Absent
Tsotsi	NM	Dr		BDS; MPH; MSc Med; PGDipInt ResEthics	F	Absent
Velaphi	S	Prof	Paediatrics	MB BCh, FCPaeds, MMed, PhD	M	Present
Wadee	R	Dr	Anatomical Pathology	MBBCh, FC Path (SA) Anat., MMed (Anatomical Pathology) DipRCPath, FAcadTM, MIAC, PhD	F	Absent
Warria	A	Dr	Social Work	D Litt et Phil (SWK)	F	Absent
Willem	P	Dr	Human Genetics	MD, PhD	F	Present
Woodiwiss	AJ	Prof	Cardiovascular Pathophysiology	BSc Physiotherapy, BSc, MSc, PhD	F	Absent

NOT AFFILIATED TO THE UNIVERSITY OF THE WITWATERSRAND

Surname	Initials	Title	Discipline/s	Academic Qualifications	Gender	Present
Barnabas	N	Ms	Civil Society Liaison Officer	Community Liaison Manager	F	Absent
Burns	I	Mr	Community Representative	MA (Hons)	M	Present
Chan	SW	Dr	Oncology	MBBCh; FCP (SA)	F	Absent
Ikalafeng	B	Dr	Governance	BSc (Hons), MSc, PhD	F	Present
Langley	G	Dr	Nursing	MSc (Nursing), PhD, MPhil (Ethics); BA (Hons) Theol	F	Present
Paruk	F	Prof	Anaesthesia	MB ChB, FCOG(SA), Crit Care(SA), PhD	F	Absent
Peter	JR	Adv	Lawyer	BCom; LLB; LLM	M	Absent

RETIRED MEMBER OCCASIONALLY CO-OPTED FOR OPINION ON A PROJECT

Stewart	A	Prof	Physiotherapy	BSc (Physio), MSc, PhD, DPE	F

Note 1: This committee has been in continuous operation since October 1966

Note 2: The large committee size is to ensure a good attendance at meetings

Note 3: A Quorum is 12 members according to the 33% of members on a committee with more than 15 members as required by SA National Guidelines.

This is to certify that the Human Research Ethics Committee: (Medical) of the University of the Witwatersrand operates according to the following guidelines of good clinical practice:

1. **ICH Harmonised Tripartite Guideline for Good Clinical Practice.**
2. **Department of Health, 2020. South African Good Clinical Practice: Clinical Trial Guidelines. Third Edition (SA GCP 2020)**
3. **Ethics in Health Research: Principles, Processes and Structures, Department of Health 2015, Second Edition**
4. **Declaration of Helsinki 2013.**

The Committee's United States Federal Wide Assurance details are:

1. **Country code SF.**
2. **FWA Number: FWA00000715.**
3. **University of the Witwatersrand: IORG0000862.**
4. **Human Research Ethics Committee: (Medical): IRB00001223.**

